

(ii) $|\zeta'(s)| \leq c_\epsilon |t|^\epsilon$, if $1 \leq \sigma$ and $|t| \geq 1$.

In particular, the proposition implies that $\zeta(1+it) = O(|t|^\epsilon)$ as $|t|$ tends to infinity,³ and the same estimate also holds for ζ' . For the proof, we use Corollary 2.6. Recall the estimate $|\delta_n(s)| \leq |s|/n^{\sigma+1}$. We also have the estimate $|\delta_n(s)| \leq 2/n^\sigma$, which follows from the expression for $\delta_n(s)$ given by (9) and the fact that $|n^{-s}| = n^{-\sigma}$ and $|x^{-s}| \leq n^{-\sigma}$ if $x \geq n$. We then combine these two estimates for $|\delta_n(s)|$ via the observation that $A = A^\delta A^{1-\delta}$, to obtain the bound

$$|\delta_n(s)| \leq \left(\frac{|s|}{n^{\sigma_0+1}} \right)^\delta \left(\frac{2}{n^{\sigma_0}} \right)^{1-\delta} \leq \frac{2|s|^\delta}{n^{\sigma_0+\delta}},$$

as long as $\delta \geq 0$. Now choose $\delta = 1 - \sigma_0 + \epsilon$ and apply the identity in Corollary 2.6. Then, with $\sigma = \operatorname{Re}(s) \geq \sigma_0$, we find

$$|\zeta(s)| \leq \left| \frac{1}{s-1} \right| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

and conclusion (i) is proved. The second conclusion is actually a consequence of the first by a slight modification of Exercise 8 in Chapter 2. For completeness we sketch the argument. By the Cauchy integral formula,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{i\theta} d\theta,$$

where the integration is taken over a circle of radius r centered at the point s . Now choose $r = \epsilon$ and observe that this circle lies in the half-plane $\operatorname{Re}(s) \geq 1 - \epsilon$, and so (ii) follows as a consequence of (i) on replacing 2ϵ by ϵ .

3 Exercises

1. Prove that

$$\Gamma(s) = \lim_{n \rightarrow \infty} \frac{n^s n!}{s(s+1) \cdots (s+n)}$$

whenever $s \neq 0, -1, -2, \dots$

[Hint: Use the product formula for $1/\Gamma$, and the definition of the Euler constant γ .]

2. Prove that

$$\prod_{n=1}^{\infty} \frac{n(n+a+b)}{(n+a)(n+b)} = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+1)}$$

³The reader should recall the O notation which was introduced at the end of Chapter 1.

whenever a and b are positive. Using the product formula for $\sin \pi s$, give another proof that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$.

3. Show that Wallis's product formula can be written as

$$\sqrt{\frac{\pi}{2}} = \lim_{n \rightarrow \infty} \frac{2^{2n}(n!)^2}{(2n+1)!} (2n+1)^{1/2}.$$

As a result, prove the following identity:

$$\Gamma(s)\Gamma(s+1/2) = \sqrt{\pi} 2^{1-2s} \Gamma(2s).$$

4. Prove that if we take

$$f(z) = \frac{1}{(1-z)^\alpha}, \quad \text{for } |z| < 1$$

(defined in terms of the principal branch of the logarithm), where α is a fixed complex number, then

$$f(z) = \sum_{n=0}^{\infty} a_n(\alpha) z^n$$

with

$$a_n(\alpha) \sim \frac{1}{\Gamma(\alpha)} n^{\alpha-1} \quad \text{as } n \rightarrow \infty.$$

5. Use the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ to prove that

$$|\Gamma(1/2 + it)| = \sqrt{\frac{2\pi}{e^{\pi t} + e^{-\pi t}}}, \quad \text{whenever } t \in \mathbb{R}.$$

6. Show that

$$1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} - \frac{1}{2} \log n \rightarrow \frac{\gamma}{2} + \log 2,$$

where γ is Euler's constant.

7. The **Beta function** is defined for $\operatorname{Re}(\alpha) > 0$ and $\operatorname{Re}(\beta) > 0$ by

$$B(\alpha, \beta) = \int_0^1 (1-t)^{\alpha-1} t^{\beta-1} dt.$$

(a) Prove that $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

(b) Show that $B(\alpha, \beta) = \int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}} du$.

[Hint: For part (a), note that

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty \int_0^\infty t^{\alpha-1} s^{\beta-1} e^{-t-s} dt ds,$$

and make the change of variables $s = ur$, $t = u(1-r)$.]

8. The Bessel functions arise in the study of spherical symmetries and the Fourier transform. See Chapter 6 in Book I. Prove that the following power series identity holds for Bessel functions of real order $\nu > -1/2$:

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1/2)\sqrt{\pi}} \int_{-1}^1 e^{ixt} (1-t^2)^{\nu-(1/2)} dt = \left(\frac{x}{2}\right)^\nu \sum_{m=0}^\infty \frac{(-1)^m \left(\frac{x^2}{4}\right)^m}{m! \Gamma(\nu+m+1)}$$

whenever $x > 0$. In particular, the Bessel function J_ν satisfies the ordinary differential equation

$$\frac{d^2 J_\nu}{dx^2} + \frac{1}{x} \frac{dJ_\nu}{dx} + \left(1 - \frac{\nu^2}{x^2}\right) J_\nu = 0.$$

[Hint: Expand the exponential e^{ixt} in a power series, and express the remaining integrals in terms of the gamma function, using Exercise 7.]

9. The hypergeometric series $F(\alpha, \beta, \gamma; z)$ was defined in Exercise 16 of Chapter 1. Show that

$$F(\alpha, \beta, \gamma; z) = \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \int_0^1 t^{\beta-1} (1-t)^{\gamma-\beta-1} (1-zt)^{-\alpha} dt.$$

Here $\alpha > 0$, $\beta > 0$, $\gamma > \beta$, and $|z| < 1$.

Show as a result that the hypergeometric function, initially defined by a power series convergent in the unit disc, can be continued analytically to the complex plane slit along the half-line $[1, \infty)$.

Note that

$$\begin{aligned} \log(1-z) &= -zF(1, 1, 2; z), \\ e^z &= \lim_{\beta \rightarrow \infty} F(1, \beta, 1; z/\beta), \\ (1-z)^{-\alpha} &= F(\alpha, 1, 1; z). \end{aligned}$$

[Hint: To prove the integral identity, expand $(1-zt)^{-\alpha}$ as a power series.]

10. An integral of the form

$$F(z) = \int_0^\infty f(t) t^{z-1} dt$$

is called a **Mellin transform**, and we shall write $\mathcal{M}(f)(z) = F(z)$. For example, the gamma function is the Mellin transform of the function e^{-t} .

(a) Prove that

$$\mathcal{M}(\cos)(z) = \int_0^\infty \cos(t)t^{z-1} dt = \Gamma(z) \cos\left(\pi \frac{z}{2}\right) \quad \text{for } 0 < \operatorname{Re}(z) < 1,$$

and

$$\mathcal{M}(\sin)(z) = \int_0^\infty \sin(t)t^{z-1} dt = \Gamma(z) \sin\left(\pi \frac{z}{2}\right) \quad \text{for } 0 < \operatorname{Re}(z) < 1.$$

(b) Show that the second of the above identities is valid in the larger strip $-1 < \operatorname{Re}(z) < 1$, and that as a consequence, one has

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2} \quad \text{and} \quad \int_0^\infty \frac{\sin x}{x^{3/2}} dx = \sqrt{2\pi}.$$

This generalizes the calculation in Exercise 2 of Chapter 2.

[Hint: For the first part, consider the integral of the function $f(w) = e^{-w}w^{z-1}$ around the contour illustrated in Figure 1. Use analytic continuation to prove the second part.]

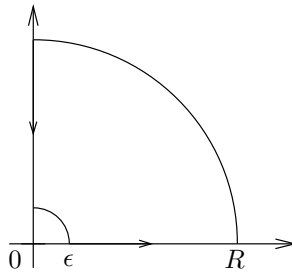


Figure 1. The contour in Exercise 10

11. Let $f(z) = e^{az}e^{-e^z}$ where $a > 0$. Observe that in the strip $\{x + iy : |y| < \pi\}$ the function $f(x + iy)$ is exponentially decreasing as $|x|$ tends to infinity. Prove that

$$\hat{f}(\xi) = \Gamma(a + i\xi), \quad \text{for all } \xi \in \mathbb{R}.$$

12. This exercise gives two simple observations about $1/\Gamma$.

- (a) Show that $1/|\Gamma(s)|$ is not $O(e^{c|s|})$ for any $c > 0$. [Hint: If $s = -k - 1/2$, where k is a positive integer, then $|1/\Gamma(s)| \geq k!/\pi$.]
- (b) Show that there is no entire function $F(s)$ with $F(s) = O(e^{c|s|})$ that has simple zeros at $s = 0, -1, -2, \dots, -n, \dots$, and that vanishes nowhere else.

13. Prove that

$$\frac{d^2 \log \Gamma(s)}{ds^2} = \sum_{n=0}^{\infty} \frac{1}{(s+n)^2}$$

whenever s is a positive number. Show that if the left-hand side is interpreted as $(\Gamma'/\Gamma)'$, then the above formula also holds for all complex numbers s with $s \neq 0, -1, -2, \dots$

14. This exercise gives an asymptotic formula for $\log n!$. A more refined asymptotic formula for $\Gamma(s)$ as $s \rightarrow \infty$ (Stirling's formula) is given in Appendix A.

- (a) Show that

$$\frac{d}{dx} \int_x^{x+1} \log \Gamma(t) dt = \log x, \quad \text{for } x > 0,$$

and as a result

$$\int_x^{x+1} \log \Gamma(t) dt = x \log x - x + c.$$

- (b) Show as a consequence that $\log \Gamma(n) \sim n \log n$ as $n \rightarrow \infty$. In fact, prove that $\log \Gamma(n) \sim n \log n + O(n)$ as $n \rightarrow \infty$. [Hint: Use the fact that $\Gamma(x)$ is monotonically increasing for all large x .]

15. Prove that for $\operatorname{Re}(s) > 1$,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

[Hint: Write $1/(e^x - 1) = \sum_{n=1}^{\infty} e^{-nx}$.]

16. Use the previous exercise to give another proof that $\zeta(s)$ is continuable in the complex plane with only singularity a simple pole at $s = 1$.

[Hint: Write

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^1 \frac{x^{s-1}}{e^x - 1} dx + \frac{1}{\Gamma(s)} \int_1^{\infty} \frac{x^{s-1}}{e^x - 1} dx.$$

The second integral defines an entire function, while

$$\int_0^1 \frac{x^{s-1}}{e^x - 1} dx = \sum_{m=0}^{\infty} \frac{B_m}{m!(s+m-1)},$$

where B_m denotes the m^{th} Bernoulli number defined by

$$\frac{x}{e^x - 1} = \sum_{m=0}^{\infty} \frac{B_m}{m!} x^m.$$

Then $B_0 = 1$, and since $z/(e^z - 1)$ is holomorphic for $|z| < 2\pi$, we must have $\limsup_{m \rightarrow \infty} |B_m/m!|^{1/m} = 1/2\pi$.]

17. Let f be an indefinitely differentiable function on $\mathbb{R}$ that has compact support, or more generally, let f belong to the Schwartz space.⁴ Consider

$$I(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} f(x) x^{-1+s} dx.$$

- (a) Observe that $I(s)$ is holomorphic for $\operatorname{Re}(s) > 0$. Prove that I has an analytic continuation as an entire function in the complex plane.
- (b) Prove that $I(0) = 0$, and more generally

$$I(-n) = (-1)^n f^{(n+1)}(0) \quad \text{for all } n \geq 0.$$

[Hint: To prove the analytic continuation, as well as the formulas in the second part, integrate by parts to show that $I(s) = \frac{(-1)^k}{\Gamma(s+k)} \int_0^{\infty} f^{(k)}(x) x^{s+k-1} dx$.]

4 Problems

1. This problem provides further estimates for ζ and ζ' near $\operatorname{Re}(s) = 1$.

- (a) Use Proposition 2.5 and its corollary to prove

$$\zeta(s) = \sum_{1 \leq n < N} n^{-s} - \frac{N^{s-1}}{s-1} + \sum_{n \geq N} \delta_n(s)$$

for every integer $N \geq 2$, whenever $\operatorname{Re}(s) > 0$.

- (b) Show that $|\zeta(1+it)| = O(\log |t|)$, as $|t| \rightarrow \infty$ by using the previous result with $N = \text{greatest integer in } |t|$.

⁴The Schwartz space on $\mathbb{R}$ is denoted by $\mathcal{S}$ and consists of all indefinitely differentiable functions f , so that f and all its derivatives decay faster than any polynomials. In other words, $\sup_{x \in \mathbb{R}} |x|^m |f^{(\ell)}(x)| < \infty$ for all integers $m, \ell \geq 0$. This space appeared in the study of the Fourier transform in Book I.